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Rayla Esquivel Zabib

Graduate Material Scientist & Engineer

Materials Science and Engineering graduate from Imperial College London, proficient in Python, extensively used for data analysis and visualisation using pandas, NumPy, and scikit-learn libraries. Experienced in C++, LaTeX, and Microsoft Office suite. Enthusiastic and result-driven with hands-on laboratory, simulation, and prototyping experience. Capable of quickly adapting and learning any skills required.

Education

Master of Engineering in Materials Science and Engineering, *Imperial College London* — Jun 2026
First Class Honours

A-levels in Maths (A*), Product Design (A*), and Chemistry (B), *LVS Ascot* — Aug 2022
Academic Scholarship, LVS Ascot — 2020 — 2022

10 A-A* GCSEs, *LVS Ascot* — Aug 2020

Skills

Specialist Software	COMSOL Multiphysics (RF Module), Fusion360, SolidWorks, KiCAD
Scientific Analysis	ImageJ, Origin, Excel, LaTeX, Python
Language Proficiencies	English, Spanish (Native Level)
Certifications	Microsoft Office Specialist (MOS) certification in Microsoft Excel (2019)
Characterisation Techniques	SQUIDs, LOM, XRD, SEM/EDX, FTIR, UV-Vis, VNA-FMR

Projects

Master's Research / Inductive Noise Spectroscopy — **Oct 2025 — Jun 2026**
Masers Research Group — *78.03%*

- Developed experimental methodology backed by literature, using Python for automating experiments, data analysis, and data visualisation
- Integrated COMSOL, MuMax3, and Python simulations for experimental predictions and optimisation.
- Trained to decant and utilise liquid nitrogen safely.
- Presented experimental findings and simulation results to research team.
- Learned SCPI and the operation of scientific equipment. Became proficient at bug fixing, and time management. Developed advanced knowledge on micromagnetic properties.

Ceramics Processing / Non-permeable Porcelain — **Oct 2024 — Apr 2025**
Processing Laboratory — *71.65%*

- Carried out an optimisation project to lower the firing temperature of porcelain while maintaining 0% water absorption.
- Worked in the preparation of the slurries and characterisation of both the raw materials and the products. Achieving a 0.1% absorption porcelain fired at $1250\,^{\circ}\mathrm{C}$
- Operated characterisation equipment including: electron microscopy (SEM), Vickers hardness testing, and general lab equipment.

FEA Heat Transfer Simulation / Jominy End Quench test — **Feb 2025 — Mar 2025**
Materials Processing — *95.00%*

- Used COMSOL to simulate a Jominy end quench test of 100Cr6 bearing steel rod to observe temperature gradient at a given time.
- Analysed data against reference CCT diagrams to predict the regions of full martensitic, full bainite, and pearlite phases.

Atomistic Simulations / Metropolis Monte Carlo — **Jan 2025 — Mar 2025**
Theory and Simulation of Materials — *80.00%*

- Carried out MMC simulation to find minimum energy landscape of a single polymer chain from different starting structures using Python.
- Analysed the effect of starting structure and temperature on number of steps needed for equilibration.
- Critically assessed the limitations and advantages of the simulation method.

Characterisation Exercise / Unknown Material Characterisation — **Jan 2024 — Mar 2024**
Materials Characterisation — *81.50%*

- Worked in a small team to identify two unknown powders and postulate their manufacturing method utilising advanced characterisation equipment.
- Became familiar sample preparation and use of X-ray diffraction (XRD), UV-Vis spectroscopy, and Scanning Electron Microscopy (SEM).
- Confident in the process of analysing, interpreting, and presenting data obtained from characterisation equipment.

Machine Learning (ML) Project / Predicting Compressive Strength of Concrete **Nov 2023 — Jan 2024**
Mathematics and Computing 2 *79.0%*

- Utilising scikit-learn and pandas libraries to adequately clean and pre-process data, and analyse the model performance across a range of metrics.
- Gained an intermediate understanding of ML pipelines and the data cleaning process.

Engineering Project / Material Property Sorting Actuator **Jan 2023 — Jun 2023**
Engineering Practice 1 *76.36%*

- Led design and manufacturing of a material property sorting actuator in a 13 person multidisciplinary engineering project.
- Managed system development from concept through fabrication and testing under time constraints. Achieving consistent accuracy with a future-proof UI made using Arduino IDE in C++.
- Edited written reports for clarity, and consistency including Gantt Charts and Risk Assessments.

Design and Technology Coursework / Educational Furniture **Sep 2020 — Jun 2022**
A-level project *A**

- Designed and manufactured a bespoke piece of furniture to aid in the development of fine motor skills for children aged 4-6, durable for outdoor storage.
- Developed an eye for detail and an appreciation for material selection.
- Gained experience in CAD/CAM, becoming familiar with 3D printing, laser cutting, CNC milling, and rapid prototyping.
- Became comfortable in the use of common hand tools and workshop equipment.

Work Experience

Global Summer School Mentor / Academic Mentor **Jun 2026 — Aug 2026**
Global Summer School *Imperial College London*

- Coordinated with staff to ensure a smooth and safe experience for all members.
- Responsible for a group of 10 students, ensuring clear communication and time keeping.

Undergraduate Teaching Assistant **Nov 2025 — Jun 2026**
Undergraduate Labs and Workshops *Imperial College London*

- Aided in the delivery of workshop sessions to 1st, 2nd, and 3rd year students.
- Clearly communicating complex topics, working through past-paper questions, and providing support with data analysis to students.

Student Shapers / Virtual Reality API Testing **May 2025 — Sep 2025**
I-Explore Programme *Imperial College London*

- Assisted in bug testing a new API for in-house Virtual Reality development tool, ViRSE.
- Oversaw redesigns to module format by gathering and analysing student feedback, becoming the first 2-term STEMM module offered.

Bar and Waiting Staff **Dec 2021 — Sep 2022**
Sodexo *Ascot Racecourse*

- Serving from draught and making signature cocktails.
- Enforcing challenge 25 and other security measures behind the bar.
- Engaging in conversation with customers and working cash and card register.

Activities and Extracurriculars

Ice Hockey **Imperial Devils Ice Hockey**

- Held the role of Equipment Manager. Responsible for keeping inventory and managing spreadsheets.
- Managing budgets, interacting with vendors for equipment and merchandise, and writing grant requests.
- Regularly started for second team, achieving silver cup at national tournament (2025) and second place gold cup (2026). Played for first team on occasion, having the opportunity to play in the London Ice Hockey Varsity.
- Aided in personal development, communication skills, and working in a team.

Volleyball **Farnborough Volleyball Club / Recreational**

- Played competitively in divisions 1 and 2 in Berkshire league until 2022.
- Regularly play recreationally after starting to play ice hockey

Competitive Esports / Counter Strike **Recreational**

- Played in a team under multiple organisations, most recently Witchdoctor Gaming. Stopped playing competitively in 2022.
- Within that time played in multiple tournaments and leagues achieving podium positions in Irish Challenger Weekend II (1st place), and CEG pro series (3rd place).
- Occasionally playing recreationally.